ENGINEERS WALLAH

CHEMISTRY

Lecture IX

Chemical Reactions and Equations



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TOPICS *to be covered*

1. Balancing a Chemical Equation
2. Types of Chemical Reactions
3. Rancidity

Knowledge Ride On

Element	No. of atoms on LHS	No. of atoms on RHS
Fe	1	3
H	2×4	2×4
O	1×4	4

Balancing a Chemical Equation

Knowledge Ride On



Types of Chemical Reactions

Knowledge Ride On



Rancidity

Knowledge Ride On



Efficiency Hacks

Glossary

Chemical equation (रासायनिक समीकरण)
Chemical reaction (रासायनिक प्रतिक्रिया)
Law of conservation of mass (द्रव्यमान के संरक्षण का नियम)
Combination reaction (संयोजन प्रतिक्रिया)
Decomposition reaction (अपघटन प्रतिक्रिया)
Displacement reaction (विस्थापन प्रतिक्रिया)
Double displacement reaction (द्विविस्थापन प्रतिक्रिया)
Redox reaction (रेडॉक्स प्रतिक्रिया)
Rancidity (विकृतगंधिता)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following is an example of a colloidal sol of liquid-in-solid?

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A Mud

B Milk

C Cheese

D Rubber

D.P.	D.M.	Type	
Liquid	Solid	Gel	Cheese, Butter, Curd etc.

Question

Fine solid particles or liquid droplets suspended in air are called _____.

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A Aerosol

B Emulsion

C Foam

D Gel

D.P.	D.M.	Type
Solid	Gas	Solid aerosol ✓
Liquid	Gas	Aerosol ✓

Concept Bites – Balancing a Chemical Equation

Need to Balance a Chemical Equation (रासायनिक समीकरण)

C-III

Follows the *law of conservation of mass* wherever written or represented → Antoinne Laurent Lavoisier (1789)

(not nuclear rxn)

In simpler chemical reactions, mass is neither created nor destroyed, i.e. mass of reactants is always equal to mass of products.

When we write it on paper we should balance

$Mg + O_2 \rightarrow MgO$

C-I → Reactant(s) → Product

C-II → $(\text{no. of atoms})_R \neq (\text{no. of atoms})_P$

Unbalanced chemical equation (Skeletal chemical eqn)

Rearrangement of atoms [no. of atoms on R = P]

$AB + CD \longrightarrow AD + CB$

↳ $(\text{mass})_R = (\text{mass})_P$

Balancing By Hit-and-Trial Method

Reactants → Products

$$a\,Fe + b\,H_2O \rightarrow c\,Fe_3O_4 + d\,H_2$$

'No changes in subscript'

	(R)	(P)
Fe	1 x 3 = 3	3
H	2 x 4 = 8	2 x 4 = 8
O	1 x 4 = 4	4

$$3\,Fe + 4\,H_2O \rightarrow Fe_3O_4 + 4\,H_2$$

Balancing By Hit-and-Trial Method

$$\underbrace{CO + 2H_2}_{R} \rightarrow \underbrace{CH_3OH}_{P}$$

	R	P
C	1	1
O	1	1
H	$2 \times 2 = 4$	4

$$1CO + 2H_2 \rightarrow 1CH_3OH$$

Concept Bites – Types of Chemical Reactions

→ Most Imp.

Combination/Synthesis Reaction (संयोजन प्रतिक्रिया)

A chemical reaction in which two or more elements or compounds (reactants) react to form a single compound (product).

It is represented as: $A + B \rightarrow AB$

Combination/Synthesis Reaction (संयोजन प्रतिक्रिया)

Element Element Compound

$$C(s) + O_2(g) \rightarrow CO_2(g)$$ + heat + light (glow)

used as a fuel in rockets

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$ + heat + light (pale blue flame)

Common name

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$$ + heat

(Quick lime) / Burnt lime (Slaked lime)

$Ca(OH)_2$ → Slaked lime, Limewash, Limewater

Combination/Synthesis Reaction (संयोजन प्रतिक्रिया)

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s) + \text{Heat energy} + \text{Light energy}$$

E E C

WHITE POWDER

EXOTHERMIC REACTION

'exit of energy' → heat energy is released → in surrounding T(↑)

heat

'White dazzling flame'

↓

To prevent damage from this (U.V. Rays)

↓

Goggles / Eye Protection

entry of energy

Heat is absorbed from surroundings → ENDOTHERMIC REACTION

T(↓)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following reaction is an example of exothermic reaction?

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A. Burning of natural gas → Heat & light energy are released

B. Respiration — Glucose + Oxygen → Carbon dioxide + Water + Energy

C. Decomposition of vegetable matter into compost

D. All are correct



Give a Thought

Why a shiny finish is formed on the walls after applying whitewash?

limewash



Give a Thought

Why a shiny finish is formed on the walls after applying whitewash?

egg shells are made from $CaCO_3$

Shiny finish

$$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

(Calcium hydroxide) — (Calcium carbonate) — ↓ evaporates

common name

- ✓ Chalk
- ✓ Limestone
- ✓ Marble

Practice and PYQ's Wallah



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Question

In which type of chemical reaction a single product is formed from two or more reactants?

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A Combination reaction

B Decomposition reaction

C Displacement reaction

D Double displacement reaction

Question

Slaked lime is chemically known as:

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A Calcium hydroxide

B Sodium hydroxide

C Calcium carbonate

D Sodium nitrate

Question

Which of the following is not a form of calcium carbonate?

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A Limewater → $Ca(OH)_2$

B Limestone ✓

C Chalk ✓

D Marble ✓

Question

A magnesium ribbons burns with a dazzling:

SSC JE 2020

A Orange flame

B Blue flame

C White flame

D Purple flame

Decomposition Reaction (अपघटन प्रतिक्रिया)

Inverse of Combination rxn

A chemical reaction in which a single compound breaks down into two or more elements or compounds when the energy is supplied in the form of heat, electricity or sunlight.

Representation

$$AB \xrightarrow{\text{heat / sunlight / electricity}} A + B$$

All decomposition rxn are endothermic.

Decomposition Reaction (अपघटन प्रतिक्रिया)

$$2FeSO_4(s) \xrightarrow{\text{Heat}} Fe_2O_3(s) + SO_2(g) + SO_3(g)$$

(Ferrous sulphate) (Ferric oxide) (Reddish-brown)

$$CaCO_3(s) \xrightarrow{\text{Heat}} CaO(s) + CO_2(g)$$

(Limestone) (Quick lime)/Burnt lime

$$2Pb(NO_3)_2(s) \xrightarrow{\text{Heat}} 2PbO(s) + 4NO_2(g) + O_2(g)$$

(Lead nitrate) (WHITE) (Lead oxide) (Yellow) (Nitrogen dioxide) (Reddish-brown) (Oxygen)

→ Thermal decomposition/Thermolysis

Thermolysis → Heat, breakdown

Decomposition Reaction (अपघटन प्रतिक्रिया)

(-ve terminal) ← Cathode

Anode (+ve terminal) →

$$2H_2O\ (l) \xrightarrow{\text{Electricity}} 2H_2\ (g) + O_2\ (g)$$

→ Electrolysis

2 : 1

PHOTOSENSITIVE MATERIAL

$$2AgCl(s) \xrightarrow{\text{Sunlight}} \underset{\text{Grey}}{2Ag(s)} + Cl_2(g)$$

(White)

greenish-yellow gas

→ Photolysis

→ Used in black & white photography

$$2AgBr(s) \xrightarrow{\text{Sunlight}} \underset{\text{Grey}}{2Ag(s)} + Br_2(g)$$

(Yellow)

→ reddish-brown gas

Practice and PYQ's Wallah



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Question

Which of the following substance is used in black and white photography?

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A. Silver chloride

B. Silver bromide

C. Silver iodide

D. All are correct

Displacement Reaction (विस्थापन प्रतिक्रिया)

A chemical reaction in which a more active or reactive element displaces a less active or reactive element from its compound. These reactions are generally found to occur in the solution.

Representation: $A + BC \longrightarrow AC + B$

Displacement Reaction (विस्थापन प्रतिक्रिया)

K ←	Kudi	Potassium
Na ←	Naal	Sodium
Ca ←	Car	Calcium
Mg ←	Maango	Magnesium
Al ←	Alto	Aluminium
Zn ←	ziskO	Zinc
Fe ←	Fir	Iron
Pb	lekar →	Lead
H ←	**Hum**	**Hydrogen**
Cu	Chole →	Copper
Hg	mathura →	Mercury
Ag	Solh →	Silver
Au	Ghumre →	Gold
Pt	Pratiek →	Platinum

Reactivity/activity of metals (↓)

H → non-metal

↓

it can behave as metals & non-metals

$Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

Blue solution — Green/light green

$Zn + FeSO_4 \longrightarrow ZnSO_4 + Fe$

Light green — Colourless solⁿ

Double Displacement Reaction (द्विविस्थापन प्रतिक्रिया)

A chemical reaction in which there is an exchange of ions, i.e. cations and anions between reactants.

Representation



Double displacement rxn
- Precipitation Rxn
- Neutralisation Rxn
- Gas forming Rxn

Double Displacement Reaction (द्विविस्थापन प्रतिक्रिया)

Type → PRECIPITATION Rxn → Precipitate → insoluble solid formed after a rxn

Colourless Solutions

$$\underset{\text{(Sodium sulphate)}}{Na_2SO_4(aq)} + \underset{\text{(Barium chloride)}}{BaCl_2(aq)} \rightarrow \underset{\text{(Barium sulphate)}}{BaSO_4(s)} + \underset{\text{(Sodium chloride)}}{2NaCl(aq)}$$

Colourless solution

White insoluble solid

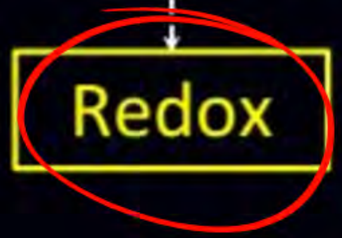
Colourless Solution

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)} + 2KNO_{3(aq)}$$

Yellow insoluble solid

Redox Reaction (रेडॉक्स प्रतिक्रिया)

A chemical reaction in which reduction and oxidation take place simultaneously. (एक साथ)





Redox Reaction (रेडॉक्स प्रतिक्रिया)

→ Oxidation

-Addition of oxygen

-Removal of hydrogen

Reduction

-Addition of hydrogen

-Removal of oxygen

Practice and PYQ's Wallah



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Question

In a reaction, what is loss of hydrogen called as?

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A Reduction

B Oxidation

C Hydroionisation

D Both reduction and hydroionisation

Question

Which among the following gets reduced in the given chemical reaction?

from left side (Reactant)

$$CuO + Zn \rightarrow Cu + ZnO$$

Addition of oxygen - OXIDATION

Removal of oxygen → Reduction

A. Zn

B. CuO ✓

C. ZnO

D. Cu

⇒ CuO $\xrightarrow[\text{to}]{\text{Reduced}}$ Cu

⇒ Zn $\xrightarrow[\text{to}]{\text{Oxidised}}$ ZnO

Oxidising Agent and Reducing Agent

(OXIDANT) (REDUCTANT)

O.A. R.A.

Oxidises others and itself gets reduced

Reduces others and itself gets ~~reduced~~ oxidised

Reduction

Oxidation

Practice and PYQ's Wallah



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Question

Identify the oxidising agent in the below chemical reaction.

$$CuO + Zn \rightarrow Cu + ZnO$$

Reducing agent

Addition of oxygen - Oxidation

Removal of oxygen - Reduction - Oxidising agent

A. Zn

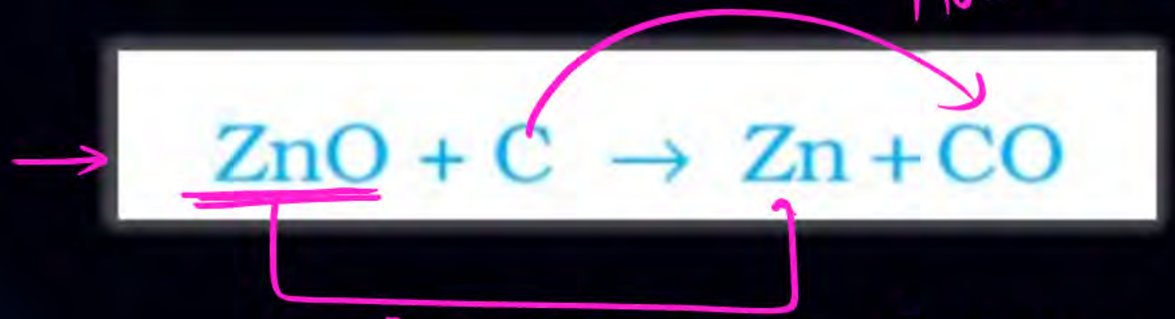
B. CuO ✓

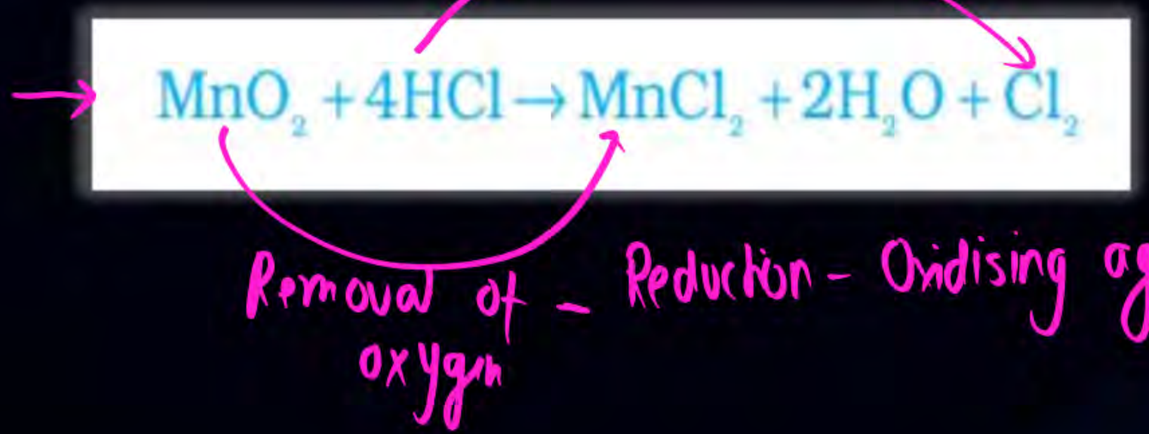
C. ZnO

D. Cu



Let's Practice





Concept Bites – Rancidity

Rancidity (विकृतगंधिता)

It is the oxidation of oil or fat-containing food items when they are kept in the open for a long time, resulting in an unpleasant odour and taste.

↳ Oxidative rancidity
(∵ of oxygen)

Some methods to prevent rancidity

(i) Addition of antioxidants: Some chemical substances are added to the fat and oil-containing substances that inhibit their oxidation. Such substances are called antioxidants.
रोकते हैं।
Examples: Vitamin C (Ascorbic acid), Vitamin E and many more

(ii) Filling nitrogen gas: In chip packets, the oxygen gas is replaced with nitrogen to prevent the oxidation of chips. Moreover, nitrogen does not react with the chips to change their odour and taste. Hence, the chips retain their original taste and odour.

(iii) Refrigeration of food items: Refrigeration lowers down the oxidation of food items and prevents a change in their odour and taste.

Efficiency Hacks!

Time-Saving Hacks

- Make a to-do list of the day. ✓
- Use social media at a fixed time for a fixed duration. ✓
- Revise through short notes. ✓




#sbsathhai
#pwsathhai
Thank
You !